

Subeshan Cobalakirouchenin

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Summary

Passionate about Machine Learning, Quantum Computing, and Full-Stack Development. Proven track record in AI hackathons with results exceeding baselines by 12-22%. Experienced in building scalable web applications, cross-platform mobile apps, and AI/ML systems. Proficient in Python, PyTorch, Flask, Flutter, and cloud technologies.

Education

Sri Manakula Vinayagar Engineering College – B.Tech in Information Technology August 2023 – Present

- CGPA: 7.86/10
- Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, OOP, Web Technologies

Amrita Vidyalayam – CBSE June 2020 – March 2023

- HSC (12th Grade): 71.0% 2023
- SSLC (10th Grade): 75.40% 2021

Projects

AI Voice Appointment Booking System | *Intelligent Automation* Oct 2025 – Dec 2025

- Developed an **AI-powered voice assistant** for booking, rescheduling, and canceling hospital appointments using natural language
- Integrated **real-time voice AI (Vapi)** with backend scheduling APIs for seamless conversational interactions
- Built a **scalable backend** using Node.js and Express.js with MySQL for appointment and user data management
- Automated end-to-end workflows using **n8n** to orchestrate voice events, API calls, and database updates
- Designed a **hospital-themed UI** with React.js and Tailwind CSS for accessibility and clarity
- **Technologies:** React.js, Tailwind CSS, Node.js, Express.js, MySQL, Vapi, n8n

Quantum-Inspired Lung Cancer Detection | *Medical AI Application* May 2025 - Sep 2025

- Designed a **hybrid quantum-classical neural network** using PennyLane + ResNet-18 for CT-scan-based lung cancer detection
- Addressed class imbalance using **advanced sampling strategies** and optimized training with early stopping
- Achieved **12–22% improvement** over baseline models in accuracy and mAP during evaluations
- Built a **Flask-based web application** with secure authentication and REST APIs for real-time diagnosis
- **Technologies:** Python, PyTorch, PennyLane, Flask, MySQL, SQLAlchemy, REST API

Technical Skills

Languages: Python, Java, C

AI/ML: TensorFlow, PennyLane, OpenCV, NumPy, Pandas

Web & Mobile: Flask, Node.js, HTML, CSS, JS, Flutter, REST APIs, Firebase

Databases & Tools: MySQL, Git, Docker, Linux(Basics)

Soft Skills: Leadership, Time Management, Teamwork, Problem Solving

Achievements

- **AI Ignite National Hackathon** – Secured **4th place** at the national level
- **LEO Summit Hackathon 2026** – Secured **1st place** in a 24-hour national hackathon
- **Unisys UIP 16** – Pre-finalist, selected among hundreds for an innovative technology solution
- **AI Agents Hackathon** – Ranked **Top 6** in agent design and development
- **Xyntra Hackathon** – Finalist, advanced to finals for a full-stack development solution

Certifications

- Skillrack - Java, C, DS, Python and SQL Basics
- IBM - Machine Learning with Python
- AWS - Data Analytics
- Microsoft - Fundamentals of Generative AI, AI in Azure
- Google Cloud - 5+ Badges in Generative AI
- NPTEL - Ethical Hacking